ASSESSING EYES

WEBER: CHAPTER 16
NURS3120: HEALTH ASSESSMENT AND PROMOTION

OBJECTIVES

1. Describe the structures and functions of the eyes
2. Differentiate between normal and abnormal findings of the eyes and vision
3. Analyze interview and physical assessment data related to the eyes and vision to formulate valid clinical judgments

EXTERNAL EYE STRUCTURES

Eyelids:

- 2 types of muscle: Striated & smooth
    - Protect the eye from foreign bodies
    - Limit the amount of light entering the eye

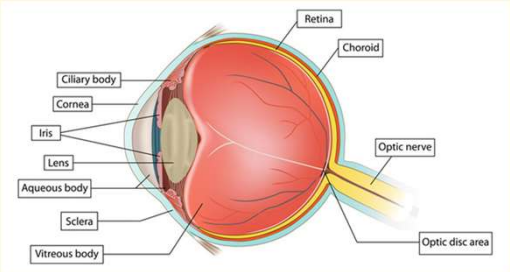
Eyelashes:

- Filter dust & dirt from air entering the eyes

Function of eyelashes: filter dust and dirt from entering the eyes
Function of the Eyelids: Protect the eye from foreign bodies; Limit the amount of light entering the eye, Distribute tears to lubricate the surface

INTERNAL EYE STRUCTURES

- **Anatomy**
  - **Sclera & cornea**
  - **Iris**
    - Circular disc of muscle
  - **Lens**
    - Bend light rays into retina
  - **Retina**
    - Receives and sends visual stimuli to the brain
  - **Optic disc**
    - Cream colored, circular area of the retina



Sclera & cornea: The external layer consists of the sclera and cornea. The sclera is a dense, protective, white covering that physically supports the internal structures of the eye. It is continuous anteriorly with the transparent cornea (the “window of the eye”). The cornea permits the entrance of light, which passes through the lens to the retina. It is well supplied with nerve endings, making it responsive to pain and touch.

Iris: is a circular disc of muscle containing pigments that determine eye color and the central aperture of the iris is called the pupil. Muscles in the iris adjust to control the pupil’s size, which controls the amount of light entering the eye. The muscle fibers of the iris also decrease the size of the pupil to accommodate for near vision and dilate the pupil when far vision is needed.

Lens: is a biconvex, transparent, avascular, encapsulated structure located posterior to the iris. The lens functions to refract (bend) light rays onto the retina.

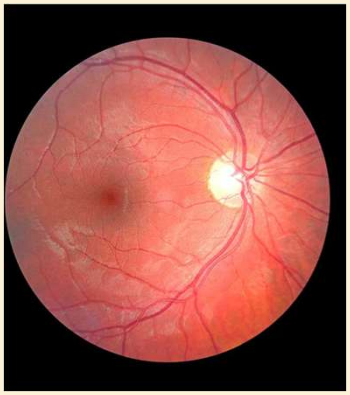
Retina: the innermost layer. It receives visual stimuli and sends it to the brain. The retina consists of numerous layers of nerve cells (“rods and cones”)

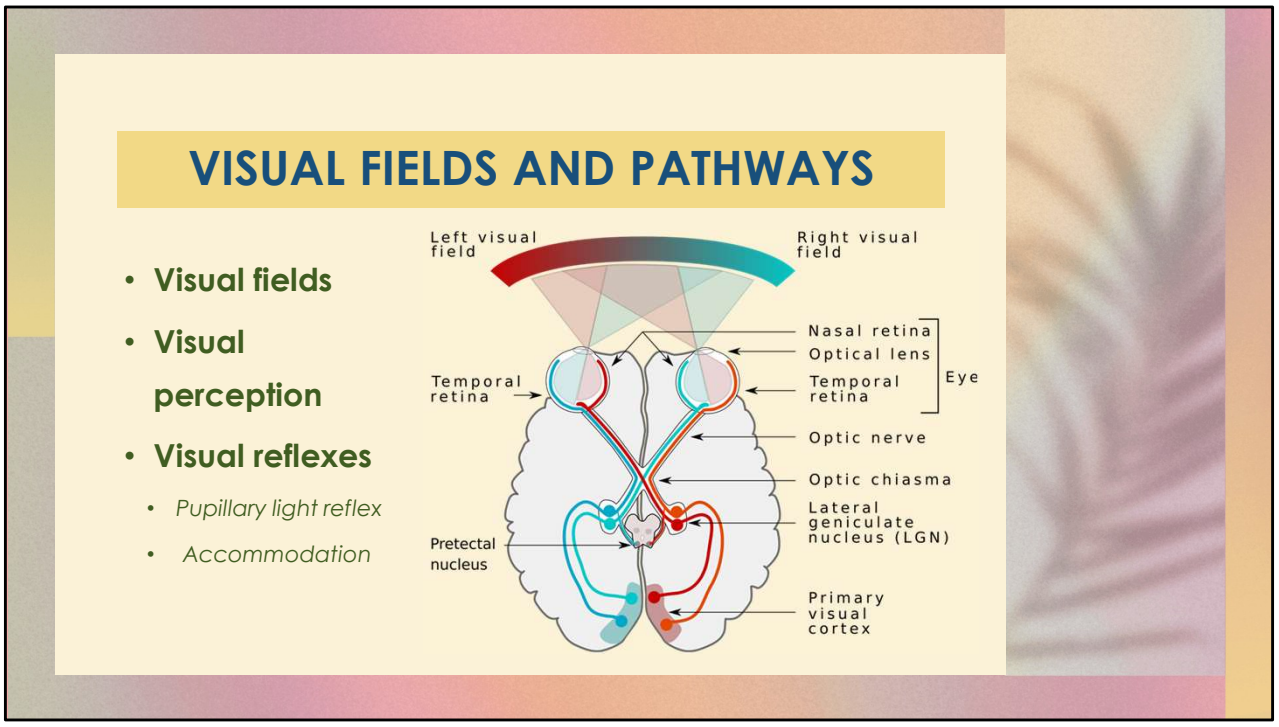
Optic Disk: cream, colored, circular area of the eye on the medial, or nasal side of the

eye, normally round or oval shape. It is where the optic nerve enters the eyeball.

OPTIC DISC

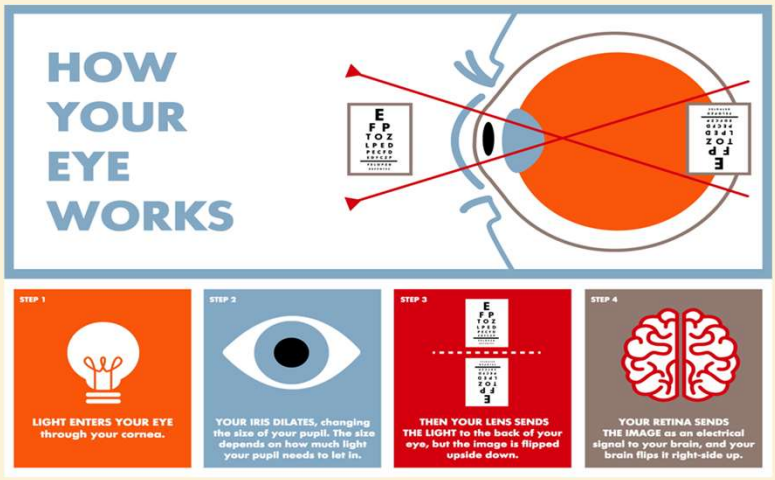
- Cream-colored, circular area located on the retina toward the medial or nasal side of the eye
- Where the optic nerve enters the eye





Visual field: what a person sees with one eye
Divided into 4 quadrants: upper temporal, lower temporal, upper nasal, and lower nasal.
The temporal quadrants of each visual field extend farther than the nasal quadrants, each eye sees a slightly different view, but their visual fields overlap quite a bit.

As a result, humans have binocular vision (two eyed vision) in which the visual cortex fuses the two slightly different images and provides depth perception or three-dimensional vision.
Visual perception occurs as light rays strike the retina where they are transformed into nerve impulses, conducted to the brain through the optic nerve and interpreted.
Visual Reflexes: the pupillary light reflex causes pupils to constrict immediately when exposed to bright light.
Accommodation is a functional reflex allowing the eyes to focus on near objects through movement of the ciliary muscles, causing an increase in the curvature of the lens. Convergence of the eyes and constriction of the pupils occur simultaneously and can be seen.


HOW YOUR EYE WORKS
STEP 1
LIGHT ENTERS YOUR EYE through your cornea.
STEP 2
YOUR IRIS DILATES, changing the size of your pupil. The size depends on how much light your pupil needs to let in.
STEP 3
THEN YOUR LENS SENDS THE LIGHT to the back of your eye, but the image is flipped upside down.
STEP 4
YOUR RETINA SENDS THE IMAGE as an electrical signal to your brain, and your brain flips it right-side up.

COLLECTING SUBJECTIVE DATA

- Vision changes
- Spots/floaters
- Blind spots
- Halos/rings
- Swelling/Redness
- Double vision
- Pain
- Discharge/itching
- Corrective lenses
- Smoker

Vision changes: Sudden vision changes are associated with acute problems such as head trauma, increased intracranial pressure. Gradual changes in vision may be related to aging, diabetes, hypertension, or neurologic disorders.
Spots/Floaters: Spots or floaters are common among clients with myopia or in clients over age 40. Normal physiological changes in the eyes that require no intervention.
Blind spots: A scotoma is a blind spot that is surrounded by either normal or slightly diminished peripheral vision. It could be caused from glaucoma, vascular spasms or a pressure on the optic nerve by a tumor or increased cranial pressure.
Halos/rings: seeing halos around lights is associated with narrow-angle glaucoma.
Double Vision: AKA Diplopia: Double vision may indicate increased intracranial pressure due to injury or a tumor
Pain/Itching: burning or itching pain is usually associated with allergies or superficial irritation. Throbbing, stabbing, or deep aching pain suggests a foreign body in the eye or changes within the eye. Most common eye disorders are not associated with actual pain. Therefore, immediately refer reports of eye pain
Redness/Swelling: is usually related to an inflammatory infection
Discharge: refers to a bacterial or viral infection
Medications: some medications have ocular side effects include an alpha 1-blocker, some antiarrhythmics, anticholinergics, anticoagulants, antimalarials, corticosteroids,

digoxin, ED medications
Glasses or Contacts: Not wearing can cause eye strain. Sunglasses: Sun exposure can cause cataracts.
Describe your typical diet: Research has linked nutrition to a decreased risk of age-related macular degeneration
Do you smoke: Smoking has been found to be strongly associated with eye diseases, doubling the chance of forming cataracts.

OLDER ADULT HEALTH HISTORY QUESTIONS



- Visual difficulty climbing stairs or driving
- Last eye exam
- Loss of central vision
- Decrease in usual activity
- Visual aids

Safety in completing ADL's!
Any history of eye problems or vision loss in your family? Is there a cloudiness in your vision?
Is the client able to administer eye drops

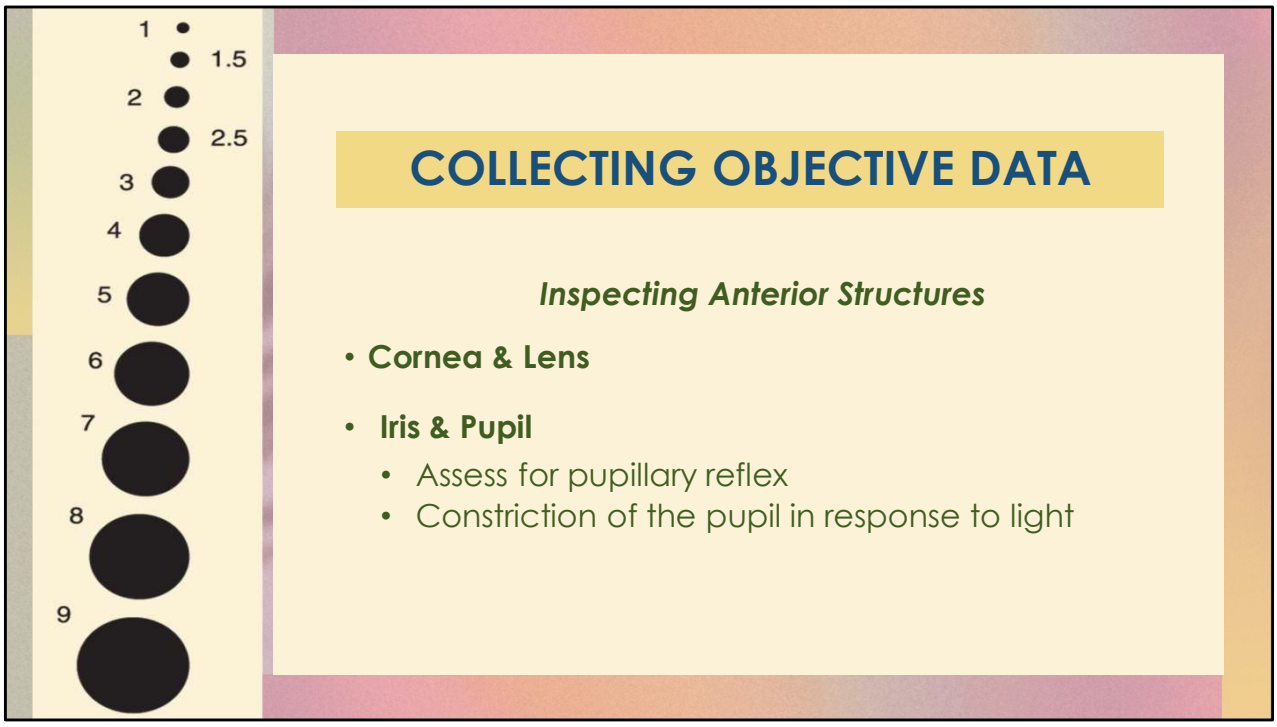
Have you been tested for glaucoma: risk factors for glaucoma? Older than 40, family history, high intraocular pressure, east Asians, farsighted past eye injury or eye surgery, DM, HTN, corticosteroids,



Inspection of external structures

When inspecting the eyelid and eyelashes observe for redness, swelling, discharge or lesions because this is not normal. Redness or crusting along the lid margins suggest seborrhea or blepharitis, an infection caused by Staphylococcus aureus. Lids should be able to close fully.

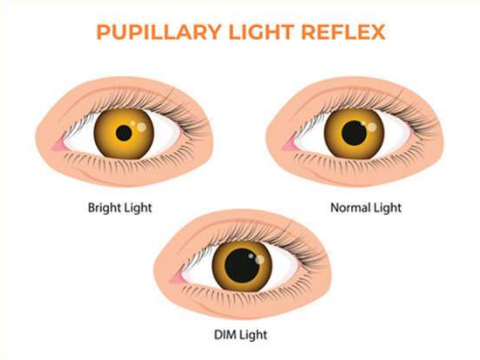
Inspect by having the patient keep the head straight while looking from side to side then up toward the ceiling. Bulbar conjunctiva is clear, moist and smooth and sclera is white: This is an excellent way to look for other problems going on with the patient….what could be abnormal? Yellow or jaundice which indicate liver problems.



When you are inspecting the shape and color of the iris and shape of pupil. Measure pupils against a gauge. Utilize the pen light and the pupil gauge to see what size the pupil is when light is shined into the eyes.

Yes this is a little subjective in information, but if your pupils are a size 9 and fixed meaning that they do not constrict when you shine a light into them, this is a serious problem. Neurological issues… if they are a size two…… first thing that comes to my mind is narcotics

PUPILLARY LIGHT REFLEX



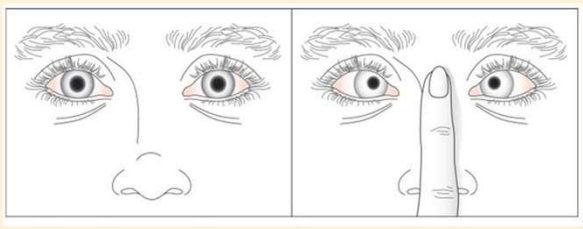


Pupillary light reflex causes pupils immediately to constrict when exposed to bright light. This can be seen as a direct reflex or as an indirect or consensual reflex, in which exposure to light in one eye results in constriction of the pupil in the opposite eye. This is a protective mechanism/reflex that your body does to prevent damage to the delicate photoreceptors by excessive light.

The pupils should ALWAYS work together. CONSENSUAL response. If they don't , this is a neurological issue.

COLLECTING OBJECTIVE DATA

- PERRLA
- Pupils
- Equal
- Round
- Reactive to
- Light &
- Accommodation



PERRLA: Pupils, Round, Reactive to Light and Accommodation
Both Pupils should constrict in response to light shone in the eyes

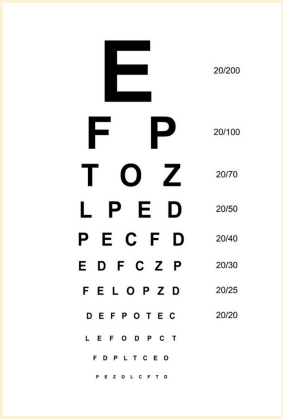
Accommodation of Pupils: Client moves focus of vision from a distant point to a near object, causing pupils to constrict
Can use a pencil or finger as on object to focus on
Normal pupillary response: constriction of the pupils and convergence of the eyes when focusing on the near object

COLLECTING OBJECTIVE DATA

Evaluating Vision

Distant visual acuity

- Snellen chart



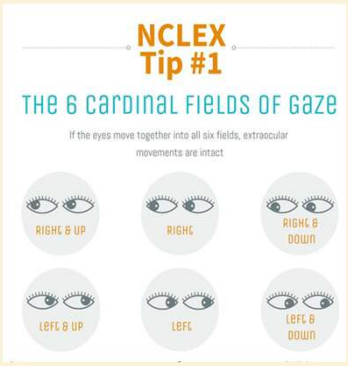
The client stands 20 ft from the chart and covers one eye.
Acuity results are recorded similar to blood pressure readings—in a manner that resembles a fraction (but in no way is interpreted as a fraction). A common example of an acuity test score is 20/20. The top, or first, number is always 20, indicating the distance from the client to the chart. The bottom, or second, number refers to the last full line the client could read.

COLLECTING OBJECTIVE DATA

Evaluating Vision

Six Cardinal Fields of Gaze

- Assesses eye muscle strength and cranial nerve function
- Have client keep head straight
- Ensure focus on the object and follow the six positions of gaze
- Assess CN III, IV, & VI

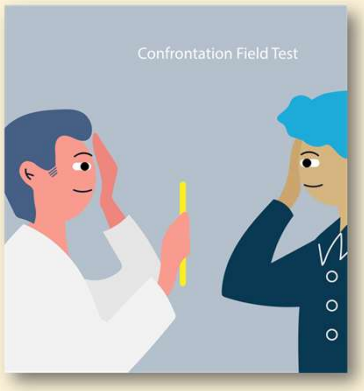


Normal: eye movement should be smooth and symmetric throughout all six directions

Abnormal: Nystagmus: Shaking movement of the eye-could be associated with an inner ear disorder, multiple sclerosis, brain lesions or narcotic use. Fine oscillating movements of the eye.

Cardinal gazes – involved CN III, IV, and VI, elicits muscle weakness or dysfunction of the cranial nerve innervating it . Top, middle, low on both sides of the eyes

COLLECTING OBJECTIVE DATA



Evaluating vision

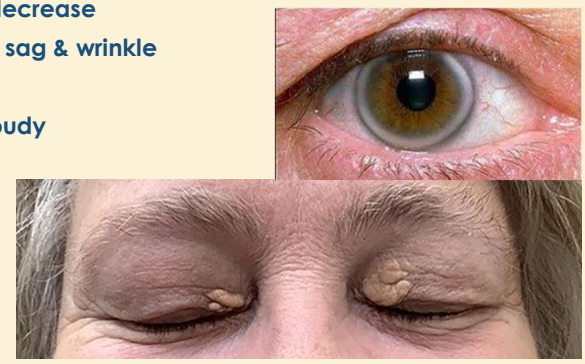
Confrontation field test

- Assesses visual fields for gross peripheral vision
- Compares client's peripheral vision from your own
- Stand 2 feet apart
- Cover opposite eyes & view the extended finger

This test is performed in the nursing profession and lot more than someone would think. Can anybody guess where this test would be performed or when? Stroke assessment, peripheral vision can be affected by a stroke….It could be a sign that a patient is having a stroke. For example, when someone is repeatedly falling it could be because they are having peripheral vision loss

OLDER ADULT CONSIDERATIONS

- Central acuity may decrease
- Skin around eyes will sag & wrinkle
- Eyes have less shine
- Cornea may look cloudy
- Arcus senilis
- Common:
  - Entropion
  - Ectropion
  - Xanthelasma



Xanthelasma, raised yellow plaques located most often near the inner canthus, is a normal variation associated with increasing age and high lipid levels.

Arcus senilis: grey/white arc around the limbus due to lipid material. Does not impair vision.

Entropion: Inverted lower lid. May cause pain or injury due to lower lashes rubbing against the conjunctiva and cornea.

Ectropion: Everted lower eyelid, resulting in exposure and drying of the conjunctiva.

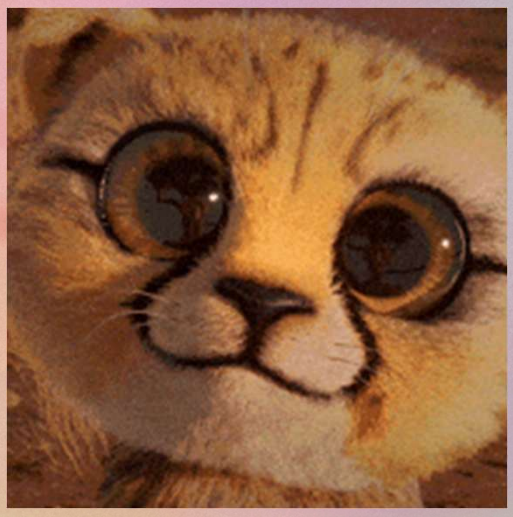
TERMINOLOGY



Photophobia
Diplopia
Myopia
Presbyopia
Scotoma
Strabismus
Nystagmus

Photophobia: Inability to tolerate light
Diplopia: Double vision
Myopia: Impaired far vision
Presbyopia: Impaired near vision that typically occurs after age 45
Scotoma: Blind spots
Strabismus: Constant malalignment of the eyes (Poor eye muscle control, farsightedness, brain or nerve problems)
Nystagmus: Oscillating movement of the eye (Inner ear disorder, multiple sclerosis, brain lesions, or narcotic use)

EYE DISEASES



MACULAR DEGENERATION

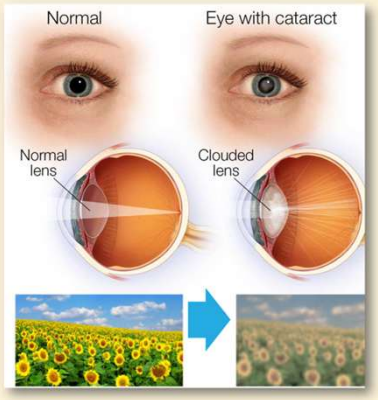
- **Aging causes damage to the macula portion of the retina**
- ***Loss of central vision***



Abnormal findings: blind spots, or blurred distorted vision
Macula processes sharp, clear straight- ahead vision

Deterioration of the retina; This is one of the major cause of visual impairment that affects the macula portion of the retina. This is one of the major leading cause of vision loss in Americans over 60 years of age.
Caucasians are more affected than any other group. Smoking, Female sex, obesity, Prolonged sun exposure, light eye color, high fat/high cholesterol diet and high sugar diet, HTN, cardiovascular disease and inactivity are all causes and risk factors to developing macular degeneration. Education to these patients is vital. Monitor diet, do not smoke, exercise regularly, wear sunglasses and regular eye exams.

CATARACTS



- **Leading cause of blindness**
- ***Cloudy film over the lens***
- **Develop slowly**
- **Most common in those >65**

Cataracts are the leading cause of blindness, often preventable, by the age of 75 more than half of the US population will have a cataract or will have had cataract surgery. If you can see…. The cataracts have a cloudy film over the lens. Most develop slowly and in those over age 65.

Symptoms: fading or yellowing of colors, sensitivity to light and glare, seeing "halos" around lights, double vision

Risk Factors: Smoking (increases risk by 2-3 times over non-smokers), Increasing age, diabetes, drinking excessive amounts of alcohol, excessive exposure to sunlight, high blood pressure, obesity, previous eye injury or inflammation, eye surgery, smoking, prolonged corticosteroid medication

GLAUCOMA

- Second leading cause of blindness
- *Loss of peripheral vision*
- Damage to the optic nerve
- Irreversible
- Caused by increased intraocular pressure

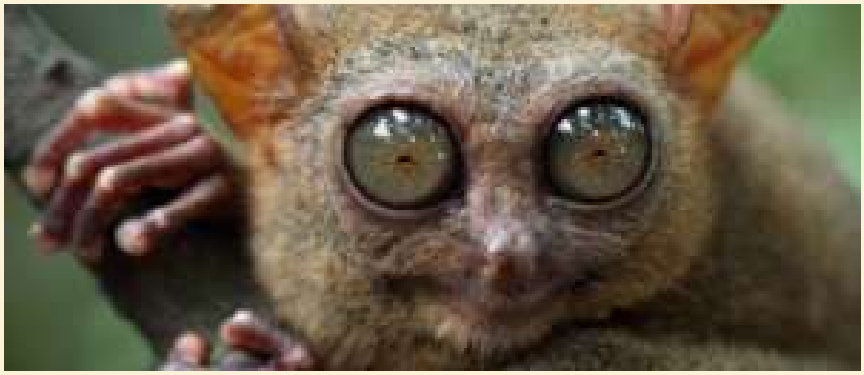


No warning signs. Glaucoma is a group of eye disease that damage the optic nerve usually caused by abnormally high blood pressure within the eye. Glaucoma is the second leading cause of blindness after cataracts. Glaucoma is irreversible. There several types of glaucoma and signs of symptoms vary on the type. Common is loss of peripheral vision and seeing halos around lights.

In Black individuals, Glaucoma occurs earlier and progresses faster.
The risk for glaucoma is 20% higher in persons of African descent.
Blindness from glaucoma is about six times more common and occurs about 10 years earlier in life in Black individuals.
After age 35, Black people should be tested for glaucoma every 1–2 years.

Screening is important to detect increased pressure early for treatment. Eye drops are used to control elevated pressure and slow progression.

ABNORMALITIES OF THE EXTERNAL EYE



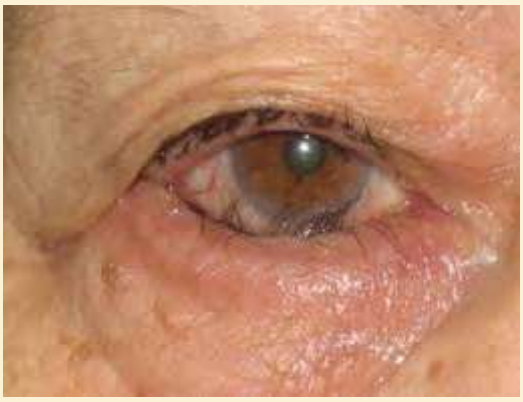


ECTROPION

OUTWARDLY TURNED LOWER LID

ENTROPION

- Inwardly turned lower lid



PTOSIS



- Drooping upper eye lid

EXOPHTHALMOS

- Protruding eyeballs and retracted eyelids
- Associated with Grave's Disease (Hyperthyroidism)



BLEPHARITIS



Inflammation of the eyelids

Staph infection of the eyelids

HORDEOLUM (STYE)



- Acute
- *Localized* staph infection of the hair follicles at the lids

CONJUNCTIVITIS

- "Pink eye"
- Generalized inflammation of the conjunctiva



JAUNDICE

- Yellowing of the sclera



Jaundice in the eyes means the whites (sclera) and sometimes skin turn yellowish due to a buildup of bilirubin, a yellow pigment from red blood cell breakdown, indicating the liver isn't processing it properly.

REVIEW QUESTION

A nurse is assessing a client's pupillary reflex. They want to chart the client's reflex was normal. What will they document?

A. Both pupils dilate in response to bright light

B. Both pupils dilate when converging to focus on the light

C. Both pupils constrict in response to bright light

D. Both pupils constrict when focusing on a distant object